Multiplayer Game Arena

Computer Networks Project#2 · Team size: **4 students**

1. Project Objective

Design and implement a **competitive multiplayer network game** using **socket programming**, built in **two versions**: one using TCP only and one using UDP only. Players must compete against one another in a shared game session. A purely cooperative game is not acceptable. Teams may choose the game concept and interface, but the focus of the project must be the **network communication and multiplayer behavior**, rather than visual complexity.

2. Architecture and Setup

- Implement a **client-server architecture** with **one server and three clients**.
- The live demonstration must use **four separate computers**: one hosting the server and one for each of the three clients.
- The application must use **network sockets** for communication. Local in-process communication or a simulated multi-client display does not satisfy this requirement.
- Build **two versions of the same game**: one that uses **TCP only** and one that uses **UDP only**. "Same game" means identical rules and features, with only the transport protocol changing.
- The **UDP only** version must meet every requirement without TCP, so teams build their own reliability and disconnection detection wherever the game needs it.
- Demonstrate a clear understanding of **TCP and UDP**, including the differences between them and how each protocol behaved in your two versions of the game.

3. Mandatory Features

Both the **TCP only** version and the **UDP only** version must implement every feature below.

Feature	Requirement
Player registration and login	Allow players to identify themselves and connect to the game platform.
Lobby and room creation	Provide a lobby where players can create game rooms.
Joining or leaving a room	Allow players to enter or exit an available room.
Game start and end	Manage game initiation, termination, and session status.
Real-time game actions	Exchange player actions and relevant game updates during competitive play.
Scores and final results	Maintain and communicate scores, rankings, and the winner at the end.
Player disconnection	Detect and handle unexpected player disconnections without crashing the system.

4. Demonstration and Submission

- **Live demonstration:** Run the server and all three clients simultaneously on four computers. Show the required features and an actual competitive game session in both the TCP only and the UDP only version.

- **Technical explanation (5 to 7 slides):** On demo day, present the socket architecture of each version and how it uses TCP or UDP. Compare the two versions using metrics your team defines, which can be qualitative, quantitative, or both (for example perceived lag, dropped updates, or desync after a lost message). Cover the friction points you faced, where each protocol performed best, and the technical differences between them. Explain how simultaneous players and disconnections are handled in each version.
- **Presentation date: Friday, November 13, 2026.**
- **Submission:** Complete source code for both versions and instructions to run each one. **No written report is required.**

5. Bonus Challenge

Cross-network multiplayer: Bonus credit is available if at least one version of the game works when the server and clients are **not all connected to the same local Wi-Fi/hotspot network**. The team must demonstrate real communication across different networks and explain the setup used.

Assessment priority: Correct socket-based networking, a well-supported TCP versus UDP comparison, reliable multiplayer operation, and a clear technical demonstration.